Summary in Graph

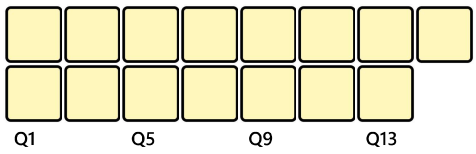
Exam Summary (GO Classes DA Test Series 2026 | Machine Learning | Topic Wise Test 4)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (10 + 5)
Total Marks:	20 (10 + 10)
Exam Duration:	45 Minutes
Time Taken:	45 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (38) out of 380 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

How many models must be trained when using k-fold cross-validation to determine which of three possible λ values $(\lambda_1, \lambda_2, \lambda_3)$ is best for ridge regression on training set with n samples (assume n is a multiple of k)?

A. $3n/k$
B. k
C. n
D. $3k$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 45min 57sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

k-fold cross-validation is equivalent to leave-one-out (LOO) cross-validation on a training set of n samples when k is equal to

A. k is not computable
B. $n-1$
C. n
D. 1

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

Suppose we are performing leave-one-out (LOO) validation and 10 -fold cross validation on a dataset of size 100,000 to pick between 4 different values of a single hyperparameter. How many times greater is the number of models that need to be trained for LOO validation versus 10 -fold cross validation?

Your Answer: | Correct Answer: 10000 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

We perform a 4-fold cross validation on 4 different hyper-parameters, the mean square error are shown in the table below. Which λ should we select?

Fold Num	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.3$	$\lambda = 0.4$	**Row max**	**Row Min**	**Row Avg**
1	80.2	84.1	70.1	91.2	91.2	70.1	83.36
2	76.8	77.3	83.3	88.8	88.8	76.8	83
3	81.5	74.5	81.6	86.5	86.5	74.5	82.12
4	79.4	75.2	79.2	85.4	85.4	75.2	80.92
Col Avg	79.475	77.775	78.55	87.975			

A. λ = 0.1
B .λ = 0.2
C .λ = 0.3
D. λ = 0.4

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Suppose we want to compute 10-Fold Cross-Validation error on 100 training examples. We need to compute error N_1 times, and the Cross-Validation error is the average of the errors. To compute each error, we need to build a model with data of size N_2, and test the model on the data of size N_3.
What are the appropriate numbers for N_1, N_2, N_3 ?

A. $N_1 = 10, N_2 = 90, N_3 = 10$
B. $N_1 = 1, N_2 = 90, N_3 = 10$
C. $N_1 = 10, N_2 = 100, N_3 = 10$

D. $N_1 = 10, N_2 = 100, N_3 = 100$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

Consider the following dataset with one real-valued input and one binary output (+ or -). We are using k-nearest neighbor (1-NN) learning with unweighted Euclidean distance to predict y for an input x.



What is the leave-one-out cross-validation error of 1-NN on this dataset? Provide your answer as the number of misclassifications.
Note: For each X, if the nearest neighbor has a different label, X would be misclassified.

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Which of the following statements are true regarding k-fold cross-validation?
1. Increasing the value of k results in more overall computation time for the cross-validation procedure
2. Using $k = n$ where n is the number of data points in the training set is the same as leave-one-out cross-validation (LOOCV).
3. Using $k = 1$ is the same the validation set method.

A. Only (1.) and (2.)
B. Only (1.) and (3.)
C. Only (2.) and (3.)
D. All three statements

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Fill in the blank to complete the pseudo-code for 'k_fold_ridge_CV', which returns the 'lambda'' from 'lambda_choices' that minimizes cross-validation MSE for ridge regression.

```
    lambda_choices = [1000, 100, 10, 1, 0.1, 0.01, 0.001]

    function k_fold_ridge_CV(k, lambda_choices):
        split data into k training and validation sets
 5.     all_errors = []

        for each lambda in lambda_choices:
            lambda_errors = []

10.         for each train_set, valid_set:
                train ridge model with lambda on train_set
                mse = _____  # (iii)
                add mse to lambda_errors

15.         add mean of lambda_errors to all_errors

        return lambda with the smallest mean error
```

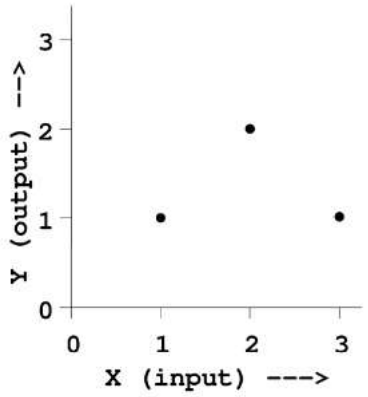
What goes in the blank (iii)?

A. Mean of squared differences between model predictions and actual values in valid_set.
B. Mean of squared differences between model predictions and actual values in train_set.
C. Mean of squared differences between the model predictions and the actual values in valid_set, plus regularization term for coefficients.
D. Mean of squared differences between model predictions and actual values in train_set, plus regularization term for coefficients.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

Consider the following data with one input and one output.



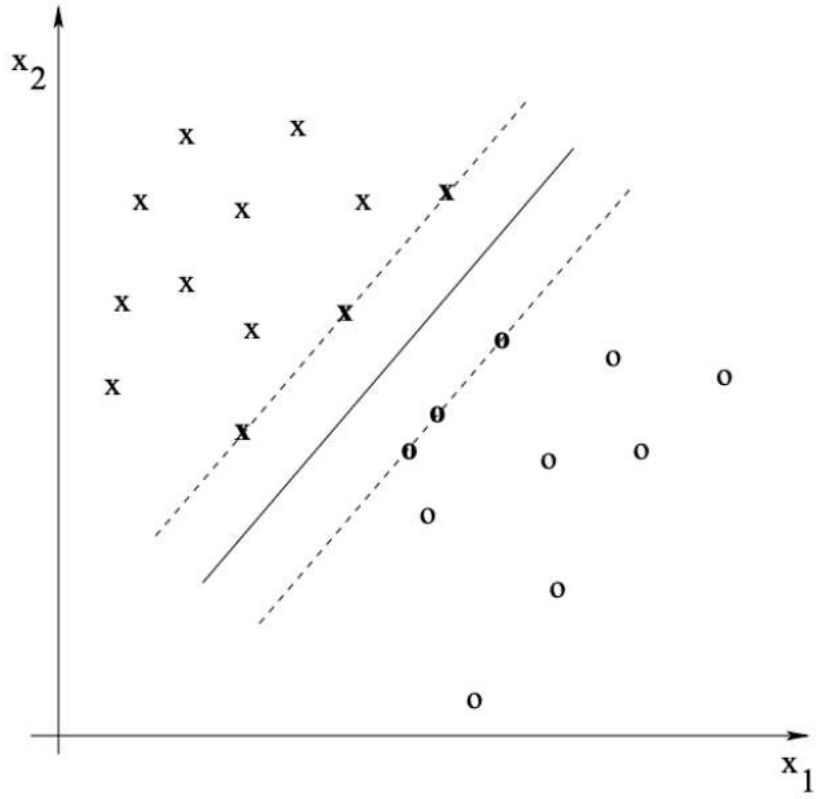
X	Y
1	1
2	2
3	1

What is the mean squared leave-one-out cross-validation (LOOCV) error of running linear regression on this data?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

What is the leave-one-out cross-validation error estimate for maximum margin separation in figure?



Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

You are fitting a model parameterized by θ using a regularized loss with a regularization parameter λ. Match each task with the appropriate type of error to use:

Tasks	Type of Errors
1. To optimize θ	A) Training Error
2. To determine the best value of λ	B) Cross-Validation Error
3. To determine the degree of polynomial features	C) Test Error
4. To evaluate the quality of your final model	

Match the tasks with the correct types of errors:

A. 1: A, 2: B, 3: C, 4: B
B. 1: B, 2: A, 3: B, 4: C
C. 1: A, 2: B, 3: B, 4: C
D. 1: C, 2: C, 3: A, 4: A

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

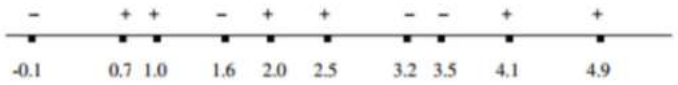
Which of the following statements are TRUE about cross validation? In this problem, the final test set is not involved in the cross validation process.

A. During the k-fold cross validation process, precisely k models are trained on different subsets of the data.
B. During the k-fold cross validation process, precisely $k-1$ models are trained on different subsets of the data.
C. During the k-fold cross validation process, we need to draw k random seeds to shuffle the data precisely k times.
D. At the end of the k-fold cross validation process, we choose hyperparameters that minimize the highest validation loss among the different splits.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Consider the following dataset with one real-valued input and one binary output (+ or -). We are using k-nearest neighbor (3-NN) learning with unweighted Euclidean distance to predict y for an input x.



What is the leave-one-out cross-validation error of 3-NN on this dataset? Provide your answer as the number of misclassifications.
Note: For each X, if the majority of the three nearest neighbors have a different label, X would be misclassified.

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

The table below shows a training set with 10 examples that is used for training a 3-nearest-neighbors classifier that uses Manhattan distance, i.e., the distance between two points at coordinates p and q is $|p-q|$. The only attribute, X, is realvalued, and the label Y has two possible classes, 0 and 1. What is the 2 -fold cross validation accuracy (percentage correct classification)? The first fold contains the first 5 examples, and the second fold contains that last 5 examples. In case of ties in distance, use the example with smallest X value as the neighbor.

X	0	1	2	3	4	5	6	7	8	9
Y	1	0	1	0	1	0	1	0	1	0

A. 0 percent
B. 20 percent
C. 40 percent
D. 60 percent

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Fill in the blanks to complete the implementation of 'create_folds', a function that takes in a value k and returns 'train_folds' and 'valid_folds' for cross-validation. Assume that ` X ' is a design matrix already defined.

```
    function create_folds(k):
        fold_size = _____  # (i)
        shuffled_idx = random_permutation of indices in X
        train_folds = []
 5.     valid_folds = []

        for i from 0 to k-1:
            train_idx = indices excluding current fold
            valid_idx = _____  # (ii)
10.         add train_idx to train_folds
            add valid_idx to valid_folds

        return train_folds, valid_folds
```

What goes in the blanks (i) and (ii)?

A. (i) Number of rows in X; (ii) Indices for the current fold from shuffled_idx.
B. (i) k; (ii) Indices for the current fold from shuffled_idx, excluding current fold.
C. (i) Number of columns in X divided by k; (ii) Indices for the current fold from shuffled_idx.
D. (i) Number of rows in X divided by k; (ii) Elements of shuffled_idx excluding train_idx.

Your Answer: | **Correct Answer: D** | Not Attempted | Time taken: 00min 00sec | Discuss

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